F79SP Session 23: Jumping chains

0.1 The jumping chain

Consider a time homogeneous Markov process with transition rate matrix $Q = (q_{ij})_{i,j\in S}$. Given $X(t) = i$, we are interested with the following questions:

- When would be the next jump of the process?
- Where would the process jump to?

When would be the next jump of the process?

- To preserve the Markov property, holding times must have an exponential distribution, since this is the only continuous random variable that has the memoryless property.
- If at any time t the process $\{X(\cdot)\}$ is in state i, the distribution of the further time to the next jump is $\text{Exp}(-q_{ii})$ (with mean $-\frac{1}{q_{ii}}$) independently of the history of the process prior to time t.

Where would the process jump to?

For any t and for any states $i, j \in S$, we have

$$\begin{aligned}\mathbf{P}(X(t+h) \neq i \,|\, X(t) = i) &= 1 - p_{ii}(h) \\ &= 1 - (1 + q_{ii}h + o(h)) \\ &= -q_{ii} + o(h) \qquad \text{as } h \to 0,\end{aligned}$$

and, for $j \neq i$,

$$\begin{aligned}\mathbf{P}(X(t+h) = j \,|\, X(t) = i) &= p_{ij}(h) \\ &= 0 + q_{ij}h + o(h) \qquad \text{as } h \to 0,\end{aligned}$$

so that

$$\begin{aligned}\mathbf{P}(X(t+h) = j \,|\, X(t) = i,\, X(t+h) \neq i) &= \frac{q_{ij}h + o(h)}{-q_{ii}h + o(h)} \\ &\to \frac{q_{ij}}{-q_{ii}} \qquad \text{as } h \to 0.\end{aligned}$$

Given at state i, when the jump occurs, it is to state $j \neq i$ with probability $-q_{ij}/q_{ii}$, independently of the time of the jump and the history of the process prior to the jump.

Constructing jumping chain

One convenient way to keep track of the movement of a continuous time Markov process will be to look at which states it jumps to, leaving the holding times for later consideration. This brings us to the notion of jumping chain, constructed as follows:

- Define

$$T_0 = 0$$
$$T_n = \text{time of } n\text{th jump of } \{X(\cdot)\} \text{ to a different state.}$$

- Conditional on $X(T_n) = i$, the distribution of the holding time is $T_{n+1} - T_n$ is $\text{Exp}(-q_{ii})$.
- Let $X_n^* = X(T_n)$, the new state of the Markov process at time T_n.

- The discrete time process $\{X_n^*\}_{n\geq 0}$ is a Markov chain, called the **jumping chain** or embedded Markov chain.
- The transition matrix of $\{X_n^*\}_{n\geq 0}$, $P^* = (p_{ij}^*)_{i,j\in S}$ is given by:

$$\text{if } q_{ii} < 0, \qquad \text{then } p_{ij}^* = -\frac{q_{ij}}{q_{ii}} \quad \text{for } j \neq i, \qquad \text{and } p_{ii}^* = 0;$$
$$\text{if } q_{ii} = 0, \qquad \text{then } p_{ij}^* = 0 \qquad \text{for } j \neq i, \qquad \text{and } p_{ii}^* = 1 \qquad (\text{state } i \text{ absorbing}).$$

Example: (Modified from T8Q1) Consider a Markov process with state space $S = \{1, 2, 3\}$ and transition rate matrix: $Q = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$. Find the transition matrix P^* of the jumping chain.

Motivation of studying the jumping chains:

- We can use the discrete time theory of the **jumping chain** $\{X_n^*\}_{n\geq 0}$ to answer many questions about the original **Markov process** $\{X(t)\}_{t\geq 0}$.
- Properties and probabilities of the Markov process which do not depend on time will be the same for the jumping chain. This includes:
  - Partitioning the state space into intercommunicating classes.
  - Classification of closed and open classes.
  - Absorption probabilities into closed classes.
  - Classification of recurrent and transient classes.
- Note, however, that the classification of **recurrent** states as **positive recurrent** or **null recurrent** may be different for the process from that for the jumping chain

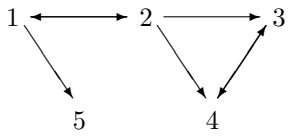
Example. Take $S = \{1, 2, 3, 4, 5\}$, and

$$Q = \begin{pmatrix} -4 & 1 & 0 & 0 & 3 \\ 2 & -6 & 3 & 1 & 0 \\ 0 & 0 & -5 & 5 & 0 \\ 0 & 0 & 4 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Then the transition matrix P^* of the jumping chain is given by

$$P^* = \begin{pmatrix} 0 & 1/4 & 0 & 0 & 3/4 \\ 1/3 & 0 & 1/2 & 1/6 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Hence we have the following *one* jump communications:



Note that $1 \to 3$ and $1 \to 4$ also.
Hence the **classes** are

$\{1, 2\}$	non-closed, hence transient
$\{3, 4\}$	closed finite, hence recurrent
$\{5\}$	closed finite (absorbing state), hence recurrent

For the **absorption probabilities**, let

$$\alpha_i = \mathbf{P}(X(\cdot) \text{ eventually absorbed in state } 5 \,|\, X(0) = i).$$

Then, letting $P^* = (p_{ij}^*)$ be the transition matrix of jumping chain, we obtain

$$\alpha_i = \sum_j p_{ij}\alpha_j \qquad\qquad j = 1, 2$$
$$\alpha_3 = \alpha_4 = 0$$
$$\alpha_5 = 1$$

Hence we obtain $\alpha_1 = 9/11$, $\alpha_2 = 3/11$ (*check*).

Further, since the set of states in nonclosed (open) classes is finite and there are just two closed classes, we have, for all i,

$$\mathbf{P}(X(\cdot) \text{ eventually absorbed in class } \{3, 4\} \,|\, X(0) = i) = 1 - \alpha_i.$$

0.2 Stationary distribution

Definition. A (probability) distribution $\boldsymbol{\pi}$ on S ($\pi_j \geq 0$ for all $j \in S$ and $\sum_{j\in S} \pi_j = 1$) is **stationary** (for the process $\{X(t)\}_{t\geq 0}$) if and only if, for all $t \geq 0$,

$$\boldsymbol{\pi} P(t) = \boldsymbol{\pi} \qquad \text{or equivalently} \qquad \sum_{i\in S} \pi_i p_{ij}(t) = \pi_j, \quad j \in S. \tag{1}$$

i.e. if the distribution of $X(0)$ is $\boldsymbol{\pi}$, then, for all $t \geq 0$, the distribution of $X(t)$ is also $\boldsymbol{\pi}$.

The following theorem shows how to obtain the stationary distribution directly from the transition rate matrix Q.

Theorem. A distribution $\boldsymbol{\pi}$ is stationary if and only if

$$\boldsymbol{\pi} Q = \mathbf{0} \qquad (\sum_{i\in S} \pi_i q_{ij} = 0 \quad \text{for all } j).$$

Proof. Will discuss later. □

Exercise: Compute stationary distribution for a MP with transition rate matrix $Q = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$.